

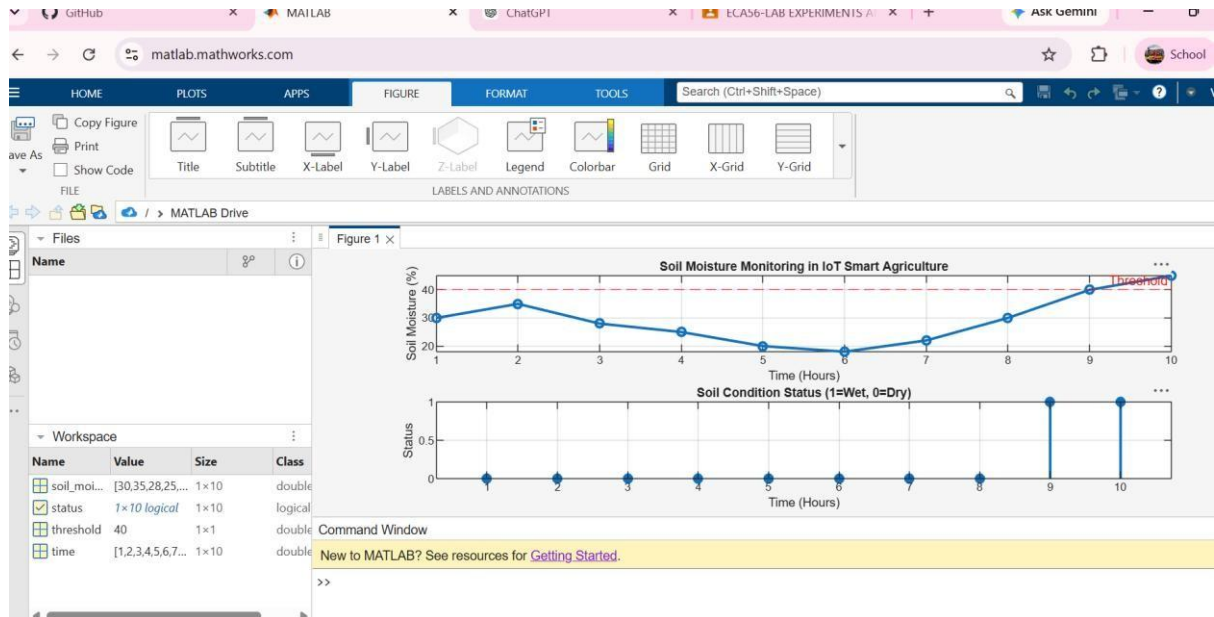
Exp 11

Obj 1

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```
clc; clear; close all;
% Simulated soil moisture data (%) time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold for irrigation
threshold = 40; figure
% ----- Graph 1: Soil Moisture Variation -----
subplot(2,1,1) plot(time, soil_moisture, '-o', 'LineWidth',
2) title('Soil Moisture Monitoring in IoT Smart
Agriculture') xlabel('Time (Hours)') ylabel('Soil Moisture
(%)') grid on hold on yline(threshold,'r--','Threshold')
% ----- Graph 2: Moisture Status (Dry/Wet Level) -----
status = soil_moisture >= threshold; subplot(2,1,2)
stem(time, status, 'filled', 'LineWidth', 2) title('Soil
Condition Status (1=Wet, 0=Dry)') xlabel('Time (Hours)')
ylabel('Status')
grid on
```



```
Obj 2 clc; clear;

close all;

% Simulated soil moisture (%) time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold level threshold
= 40;

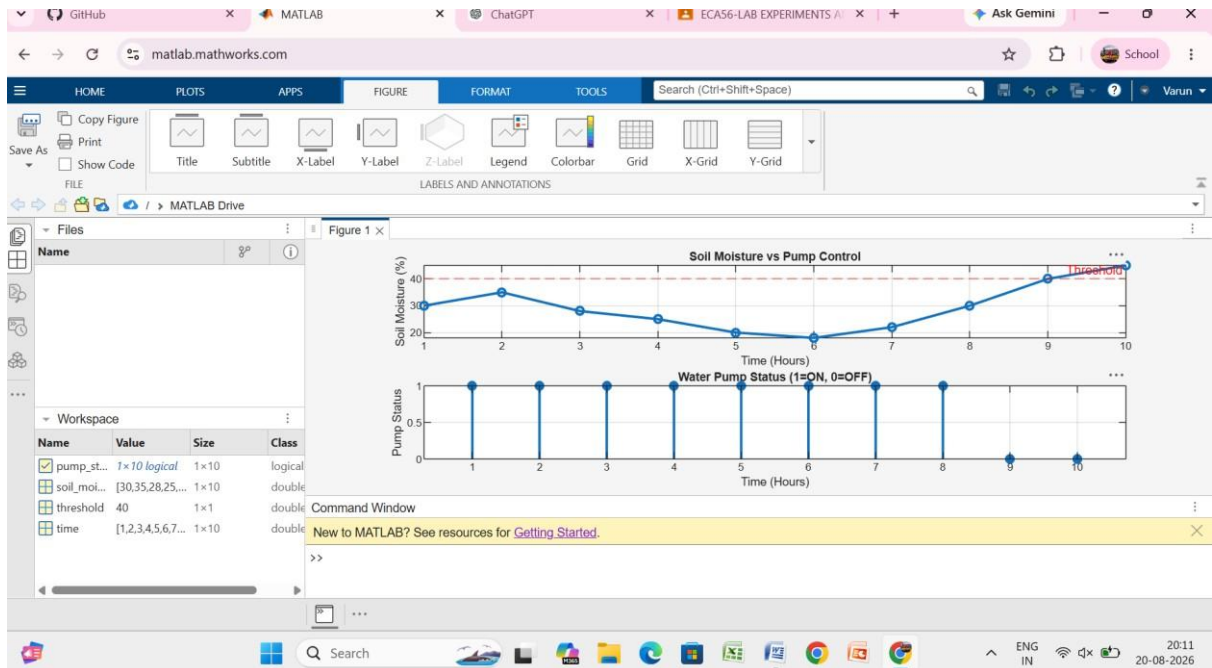
% Pump control logic (1 = ON, 0 = OFF)
pump_status = soil_moisture < threshold; figure

% ----- Graph 1: Soil Moisture vs Pump Status -----
subplot(2,1,1) plot(time, soil_moisture, '-
o','LineWidth',2) title('Soil Moisture vs Pump Control')
xlabel('Time (Hours)') ylabel('Soil Moisture (%)') grid
on hold on
yline(threshold,'r--','Threshold')

% ----- Graph 2: Pump Status (ON/OFF) -----
subplot(2,1,2) stem(time, pump_status,
'filled','LineWidth',2) title('Water Pump Status
```

(1=ON, 0=OFF)' xlabel('Time (Hours)')

ylabel('Pump Status') grid on



Obj 3

clc; clear;

close all;

% Simulated soil moisture data (%) time = 1:10;

soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold level threshold = 40;

% Irrigation efficiency calculation (%) efficiency

= (soil_moisture ./ threshold) * 100;

% Limit efficiency to 100% efficiency(efficiency

> 100) = 100;

figure;

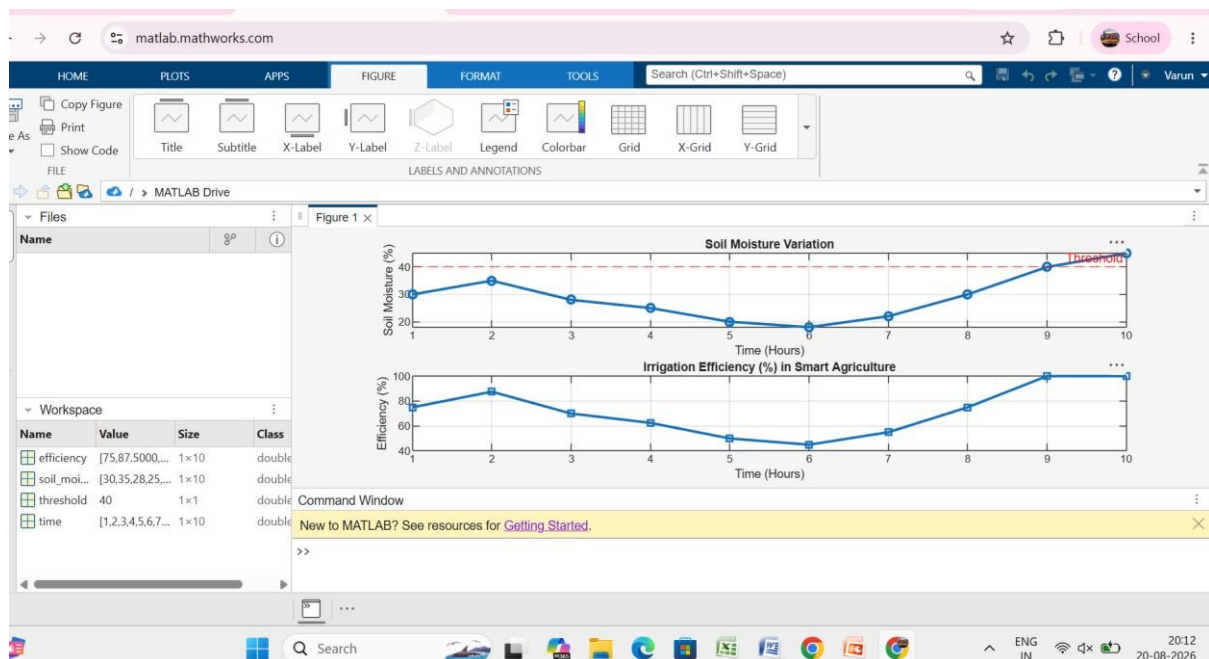
```
% ----- Graph 1: Soil Moisture vs Efficiency ----- subplot(2,1,1);
```

```
plot(time, soil_moisture, '-o', 'LineWidth', 2);  
title('Soil Moisture Variation'); xlabel('Time  
(Hours)'); ylabel('Soil Moisture (%)'); grid  
on;
```

```
hold on; yline(threshold, 'r--',  
'Threshold');
```

```
% ----- Graph 2: Irrigation Efficiency -----  
subplot(2,1,2);
```

```
plot(time, efficiency, '-s', 'LineWidth', 2); title('Irrigation  
Efficiency (%) in Smart Agriculture'); xlabel('Time  
(Hours)');  
ylabel('Efficiency (%)'); grid  
on;
```



EXPERIMENT 12

OBJ 1

```
clc; clear;
```

```
close all;
```

```
% SNR values (dB)
```

```
SNR = 0:5:30;
```

```
% Simulated Throughput (Mbps)
```

```
NB_IoT = 0.05*(1-exp(-SNR/8)); LTE_M =
```

```
1.2*(1-exp(-SNR/8)); disp('SNR (dB) NB-
```

```
IoT(Mbps) LTE-M(Mbps)') disp([SNR'
```

```
NB_IoT' LTE_M']) figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1) plot(SNR,NB_IoT,'-
```

```
o','LineWidth',2) title('NB-IoT
```

```
Throughput vs SNR') xlabel('Signal-
```

```
to-Noise Ratio (dB)')
```

```
ylabel('Throughput (Mbps)') grid on
```

```
% ----- Graph 2 ----- subplot(2,1,2)
```

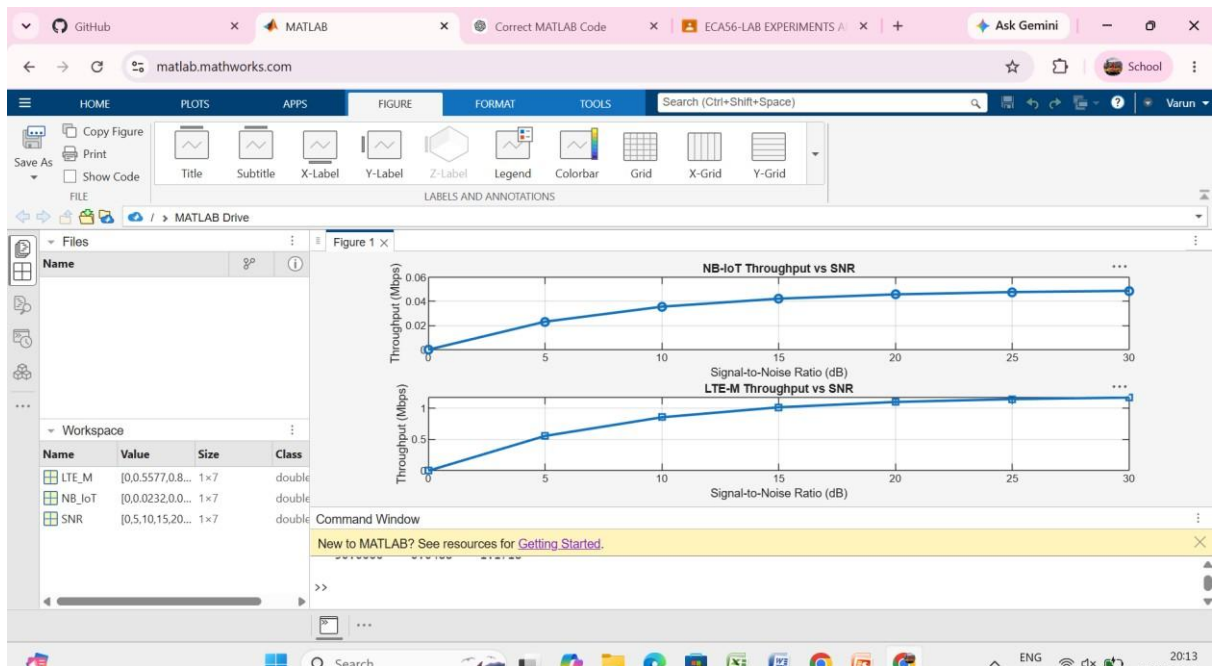
```
plot(SNR,LTE_M,'-s','LineWidth',2)
```

```
title('LTE-M Throughput vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Throughput (Mbps)') grid
```

```
on
```



OBJ 2

clc; clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Latency (ms)

NB_IoT = [180 160 145 130 120 110 100]; LTE_M = [90

80 70 60 50 45 40]; disp('SNR(dB) NB-IoT Latency(ms)

LTE-M Latency(ms)') disp([SNR' NB_IoT' LTE_M'])

figure

% ----- Graph 1 ----- subplot(2,1,1)

plot(SNR,NB_IoT,'-o','LineWidth',2)

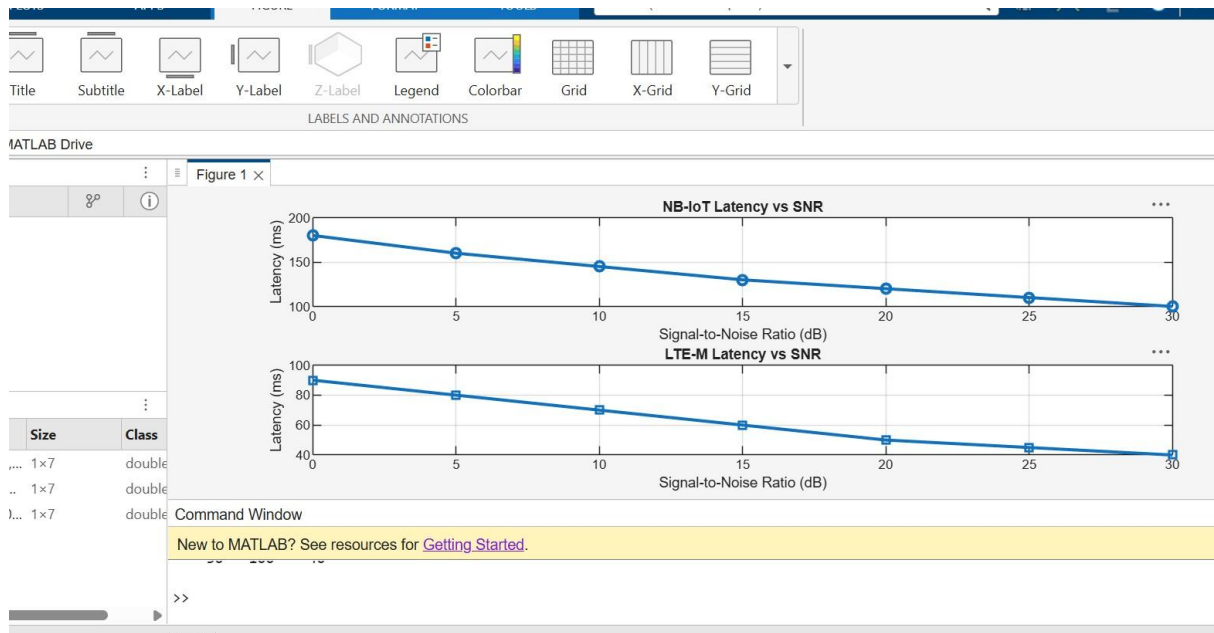
title('NB-IoT Latency vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Latency (ms)') grid on

% ----- Graph 2 -----

```
subplot(2,1,2) plot(SNR,LTE_M,'-
s','LineWidth',2) title('LTE-M
Latency vs SNR') xlabel('Signal-to-
Noise Ratio (dB)') ylabel('Latency
(ms)') grid on
```



OBJ 3

```
clc; clear;
```

```
close all;
```

```
% SNR values (dB)
```

```
SNR = 0:5:30;
```

```
% Simulated Packet Delivery Ratio (%)
```

```
NB_IoT = [82 86 89 92 95 97 99]; LTE_M = [88
```

```
91 94 96 98 99 100]; disp('SNR(dB) NB-IoT
```

```
PDR(%) LTE-M PDR(%)) disp([SNR' NB_IoT'
```

```
LTE_M'])
```

```
figure
```

```

% ----- Graph 1 ----- subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on

% ----- Graph 2 ----- subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)') grid on

```

